

HPV Vaccination Uptake and Determinants in Nigeria

Prepared by: Fauziyyah Abdulsalam

Date: June 2026

Executive Summary

Human Papillomavirus (HPV) is a leading cause of cervical cancer, posing a significant threat to women's health globally. In Nigeria, the National HPV Vaccination Program targets girls aged 9-14 to ensure early protection. This report provides a comprehensive, humanized analysis of the socioeconomic and geographic determinants influencing vaccine uptake, based on a dataset of 4,600 respondents. By understanding the barriers and drivers of vaccination, public health stakeholders can design more effective, empathetic interventions.

1. Introduction: The Human Context of Data

Behind every data point in this analysis is a young girl, a concerned parent, and a community striving for better health outcomes. The dataset focuses exclusively on females aged 9-14, aligning with clinical priorities for cervical cancer prevention.

Our analysis seeks to answer critical questions:

- Why do some families complete the vaccination series while others do not?
- How does geographic location impact access to healthcare?
- What role does parental awareness and knowledge play in the decision-making process?

By addressing these questions, we aim to translate raw data into actionable, human-centric strategies.

2. The Current Vaccination Landscape

Understanding the baseline is the first step toward improvement. The dataset reveals a population in transition, with significant progress made but critical gaps remaining.

Vaccination Status	Percentage of Population	Insight
Full Coverage	39.4%	These respondents have completed the full recommended dosage, representing the success of current outreach efforts.
Partial Uptake	38.7%	A significant portion has started but not completed the series, indicating a need for follow-up and reminder systems.
Unvaccinated	21.9%	This group represents the most vulnerable population and the primary target for future interventions.

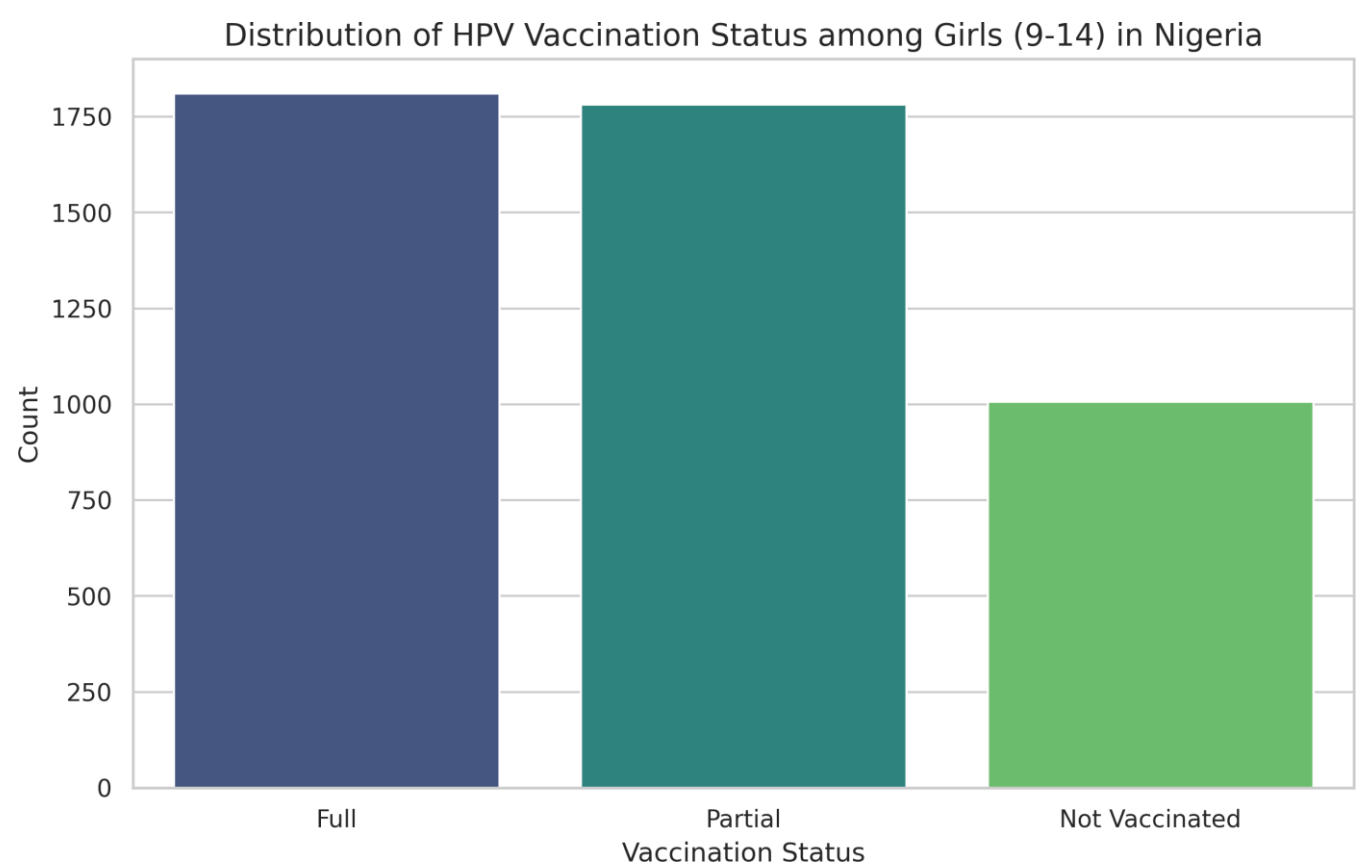


Figure 1: Distribution of HPV Vaccination Status among Girls (9-14) in Nigeria.

3. Regional Disparities: Geography as Destiny?

Healthcare access in Nigeria is not uniform. Our analysis highlights stark contrasts between different geographic regions, emphasizing the need for localized strategies.

- ◆ **Northern Regions:** States like Kano (92.6%) and Kebbi (91.5%) show high initial uptake rates. However, this often masks challenges in completing the full dosage.
- ◆ **Southern Regions:** States such as Delta (64.4%) and Osun (63.6%) exhibit lower overall vaccination rates, pointing to distinct barriers in awareness or accessibility.

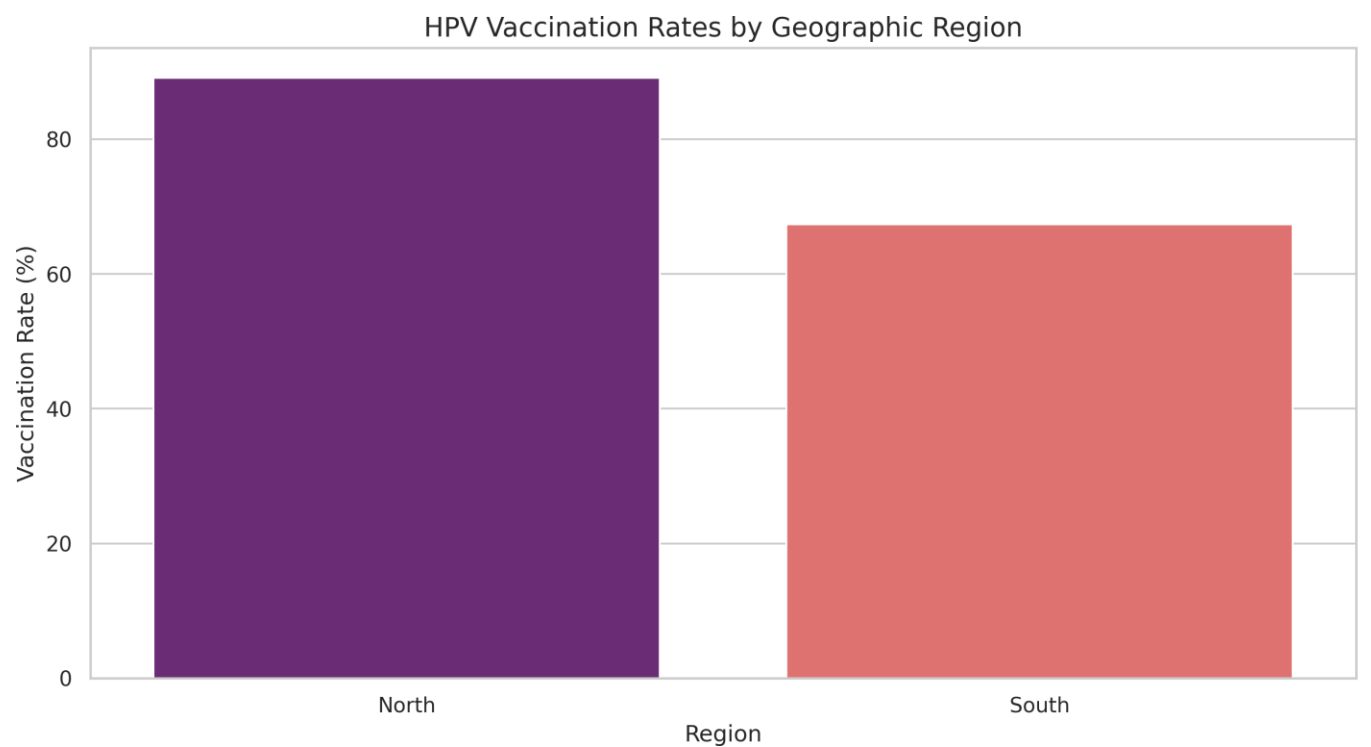


Figure 2: HPV Vaccination Rates by Geographic Region.

Strategic Implication: A one-size-fits-all approach will not work. Northern regions may require focus on dosage completion, while Southern regions need foundational awareness campaigns.

4. The Power of Knowledge and Awareness

Education is arguably the most effective vaccine against hesitancy. Our analysis demonstrates a clear, undeniable link between parental knowledge and clinical action.

1. Knowledge Scores and Vaccination

Respondents with higher knowledge scores (measured on a scale of 0-10) are significantly more likely to have their children fully vaccinated. Conversely, the “Not Vaccinated” group consistently

shows lower median knowledge scores.

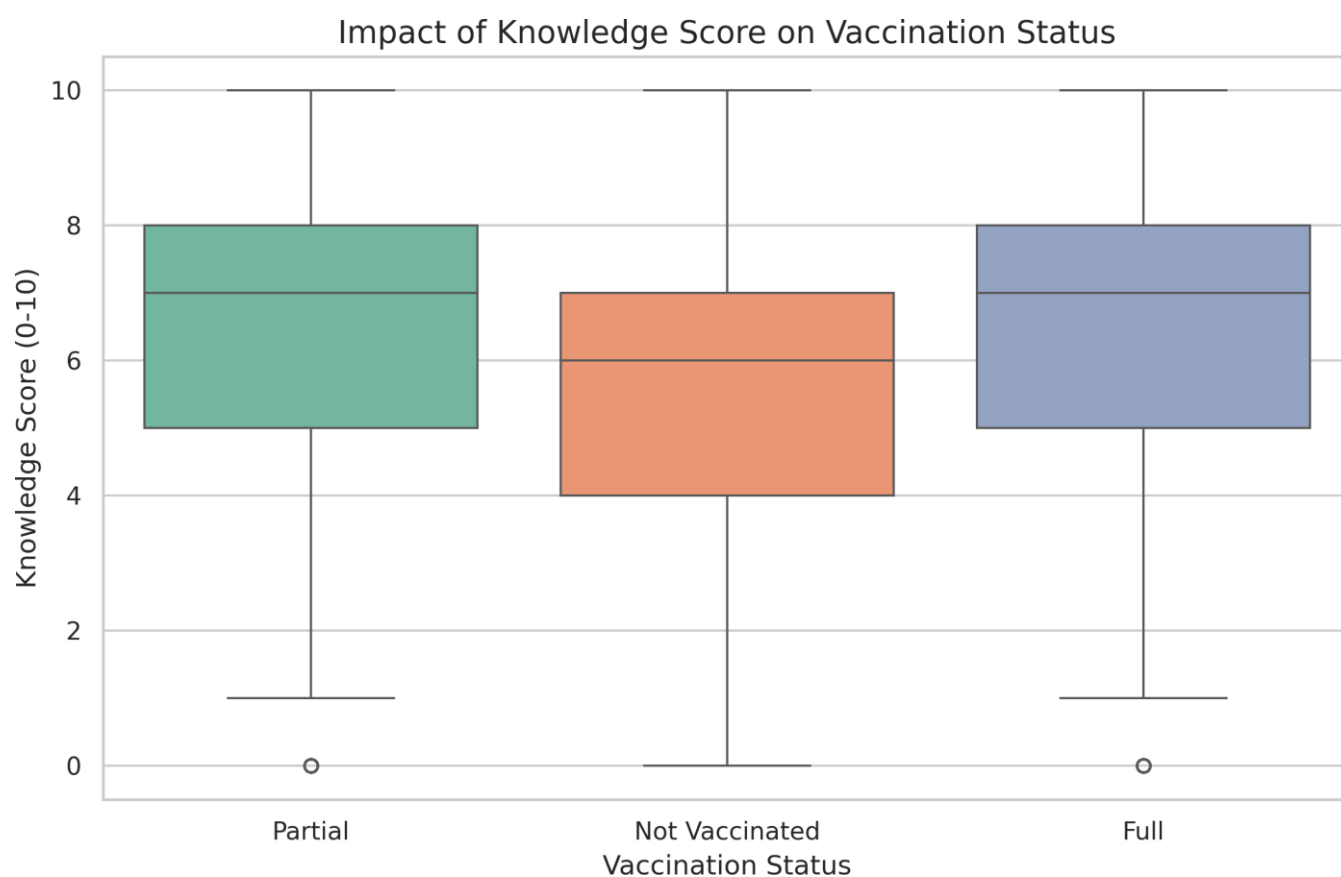


Figure 3: Impact of Knowledge Score on Vaccination Status.

4.2 Awareness Drives Willingness

Parental awareness is the primary catalyst for vaccination willingness. Parents with “High Awareness” show a significantly higher propensity to vaccinate. When awareness is low, willingness drops sharply, creating a hesitancy gap.

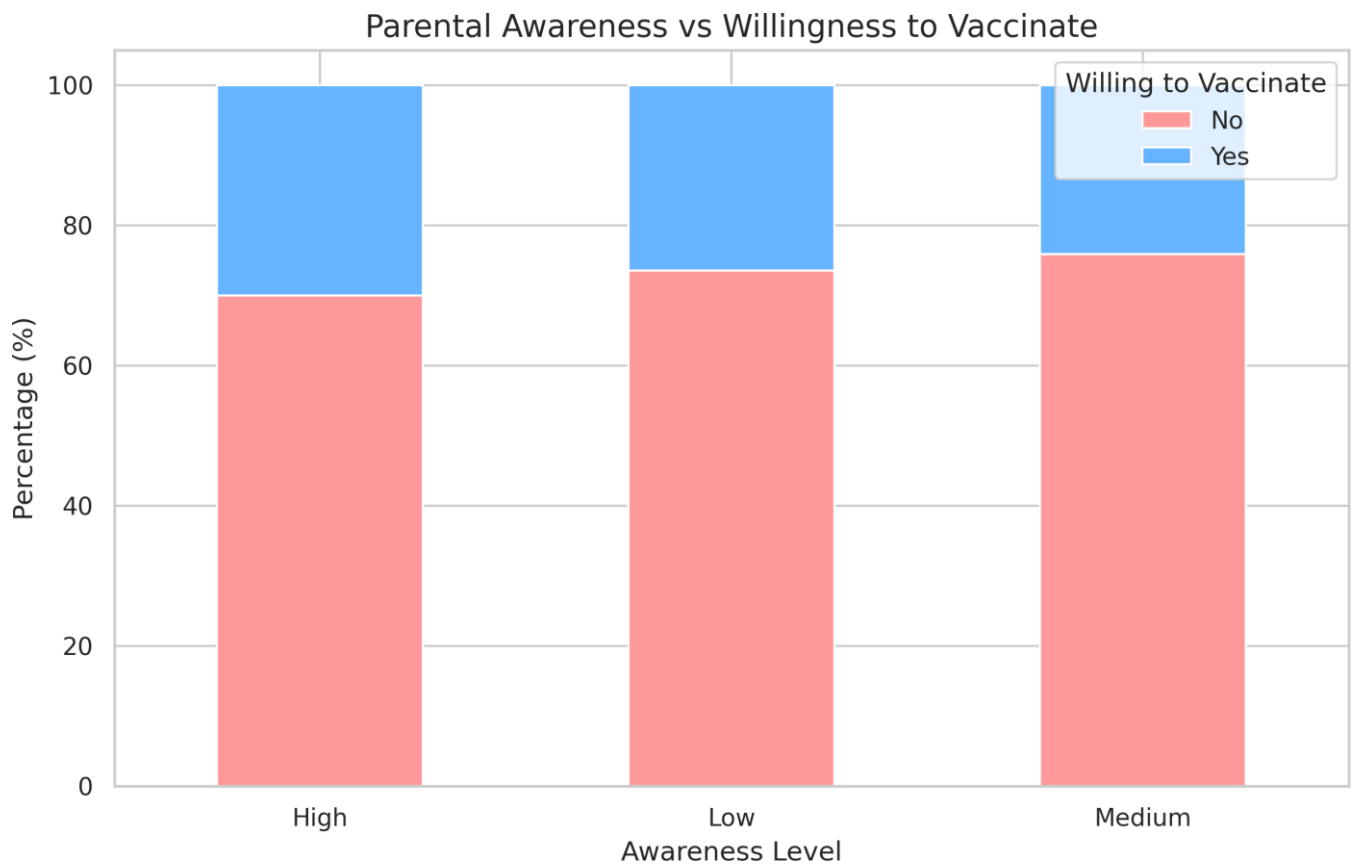


Figure 4: Parental Awareness vs Willingness to Vaccinate.

5. Accessibility: Bridging the Physical Divide

Even with high awareness and willingness, physical barriers can prevent vaccination. The accessibility score (a composite metric of distance and facility availability) varies widely across the population.

- ♦ **Rural Challenges:** Distance to health facilities remains a primary barrier in rural areas, where the average distance is significantly higher than in urban centers.
- ♦ **Urban Advantage:** Urban populations generally benefit from closer proximity to clinics, though socioeconomic factors still play a role.

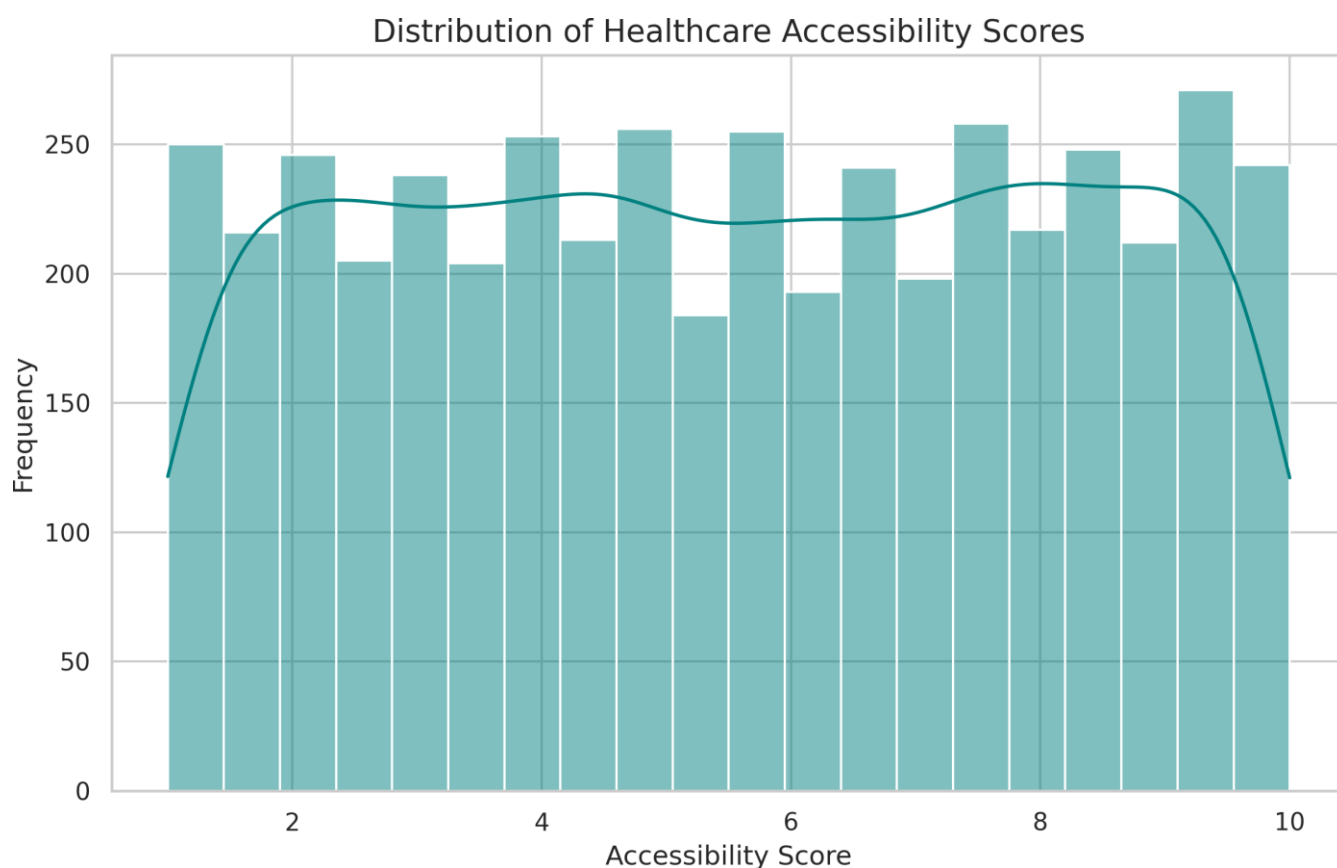


Figure 5: Distribution of Healthcare Accessibility Scores.

6. Conclusion and Strategic Recommendations

The data tells a clear story: while progress has been made, achieving universal HPV vaccination coverage in Nigeria requires targeted, empathetic, and localized interventions.

Based on our findings, we recommend the following strategic actions:

- 1. Targeted Educational Campaigns:** Deploy community-led initiatives to debunk myths and improve baseline knowledge scores, particularly in regions with high hesitancy.
- 2. Localized Infrastructure Support:** Implement mobile clinics and community outreach programs to bridge the physical accessibility gap in rural and underserved areas.
- 3. Completion Follow-up Systems:** Develop robust reminder systems (e.g., SMS alerts, community health worker visits) to ensure that the 38.7% of girls with “Partial” vaccinations complete their full dosage.
- 4. Empower Local Leaders:** Utilize religious and community leaders to advocate for vaccination, leveraging their influence to build trust in the healthcare system.

By humanizing the data and focusing on the individuals behind the numbers, we can build a healthier, more resilient future for Nigeria's young women.
